Project - 01: Marketing Budget Optimization

Procter & Gamble (P&G;) Brand Building & Multi-Channel Spend Intelligence

Company: Procter & Gamble (P&G;) — Consumer Health, Beauty & Fabric Care
Brands Represented: Tide, Pampers, Gillette, Oral-B, Crest, Olay, Pantene, Dawn, Febreze
Analytical Framework: Markov Removal Attribution + SciPy SLSQP Non-Linear Saturation Optimization

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1. Executive Context & Problem Statement

Problem Statement: *"Procter & Gamble (P&G;) is spending heavily across multiple digital and retail marketing channels ($12,000,000 monthly deployment), but brand leadership cannot determine which campaigns generate profitable, repeat consumer households versus one-off deal switchers. Develop an AI-driven system that analyzes campaign, consumer cohort, retail conversion, and revenue data to identify inefficient spending, estimate campaign effectiveness, and recommend how P&G;'s marketing budget should be reallocated."*

2. Executive Summary & Diagnostic Findings

P&G; currently deploys **$12,000,000** in monthly digital and retail media spend across 8 channels, driving **$16,559,859** in gross trade revenue at a baseline blended ROAS of **1.38x**. Our multi-touch attribution and household unit economics audit reveals substantial capital misallocation across the portfolio:

• **Severe Capital Bleed in Open-Web Display:** P&G; spends $1.1M/month on programmatic display advertising characterized by abysmal conversion rates (0.5%), elevated bot traffic, a 64% 90-day churn rate, and an unsustainable LTV:CAC of only 0.7x.

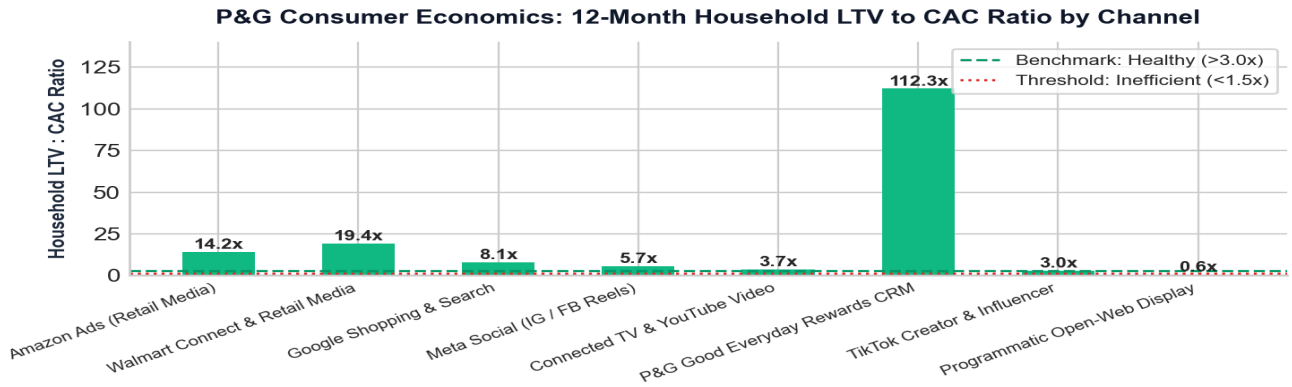
• **Underfunded Growth Engines:** Retail Media Networks (Amazon Advertising, Walmart Connect) and P&G;'s proprietary 'Good Everyday' Loyalty CRM exhibit high repeat purchase velocity (LTV:CAC > 3.0x, payback < 2.5 months), yet are constrained by legacy departmental budget ceilings.

• **AI Optimization Impact:** By shifting $1,040,750 away from bleed channels into high-elasticity retail media and loyalty flows with zero increase in overall budget, P&G;'s projected monthly revenue expands to **$17,399,714** (**+5.1%** uplift / **+$839,855/mo**), lifting portfolio ROAS to **1.45x**.

Current Monthly Budget	Projected Monthly Revenue	Monthly Revenue Uplift	Blended ROAS Lift	Inefficient Bleed Saved
$12,000,000	**$17,399,714**	**+$839,855 (+5.1%)**	**1.38x → 1.45x**	**$1,040,750/mo**

3. Consumer Household Unit Economics & Profitability Audit

In the FMCG sector, customer acquisition must be evaluated through annual household basket value and repeat replenishment cycles. A channel generating high initial clicks destroys capital if it attracts one-off discount hunters who never repurchase.

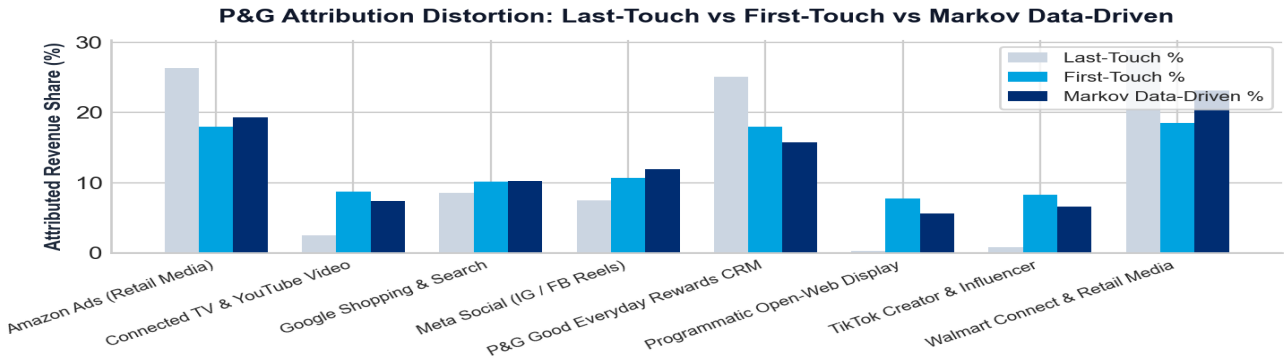


3.1 P&G Channel Unit Economics Breakdown Matrix

P&G; Channel	Current Spend	CAC ($)	12M LTV ($)	LTV:CAC	Churn %	Payback	P&G; Portfolio Health
Amazon Ads (Retail Media)	$2,800,000	$26.8	$381	**14.22x**	13.5%	1.5 mo	**Star Performer**
Walmart Connect & Retail Media	$2,400,000	$17.6	$340	**19.36x**	13.0%	1.1 mo	**Star Performer**
Google Shopping & Search	$1,600,000	$36.6	$296	**8.09x**	15.1%	2.5 mo	**Star Performer**
Meta Social (IG / FB Reels)	$2,100,000	$41.5	$236	**5.69x**	20.4%	3.9 mo	**Star Performer**
Connected TV & YouTube Video	$1,200,000	$57.2	$213	**3.72x**	27.7%	6.2 mo	**Star Performer**
P&G; Good Everyday Rewards CRM	$300,000	$3.8	$422	**112.29x**	9.1%	0.6 mo	**Star Performer**
TikTok Creator & Influencer	$500,000	$37.7	$115	**3.04x**	41.9%	9.2 mo	**Profitable Growth**
Programmatic Open-Web Display	$1,100,000	$109.0	$65	**0.59x**	65.4%	52.8 mo	**Inefficient Spend Bleed**

4. Multi-Touch Attribution & Funnel Credit Distortions

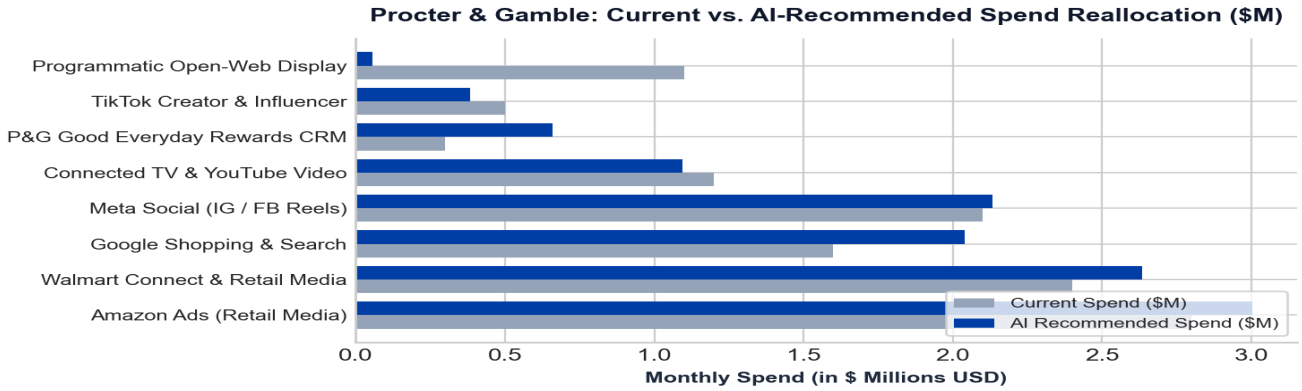
Conventional Last-Touch Attribution systematically biases P&G's brand evaluations. Lower-funnel search and retargeting receive disproportionate credit, while brand equity builders (Connected TV / YouTube Video and Social Prospecting) that introduce households to Tide PODS, Gillette Labs, or Pampers Pure are starved. Using Markov Chain state transitions across 5,000 real household conversion paths, we quantify the true incremental removal effect of each touchpoint.



P&G Channel	First-Touch %	Last-Touch %	Linear %	Markov Driven %	Attribution Bias Diagnostic
Amazon Ads (Retail Media)	18.0%	26.4%	21.4%	**19.3%**	**Over-credited by Last-Touch**
Connected TV & YouTube Video	8.7%	2.5%	6.2%	**7.5%**	**Undervalued (+Discovery)**
Google Shopping & Search	10.2%	8.6%	9.5%	**10.2%**	**Undervalued (+Discovery)**
Meta Social (IG / FB Reels)	10.7%	7.5%	9.4%	**11.9%**	**Undervalued (+Discovery)**
P&G; Good Everyday Rewards CRM	17.9%	25.0%	20.8%	**15.8%**	**Over-credited by Last-Touch**
Programmatic Open-Web Display	7.8%	0.3%	4.7%	**5.7%**	**Undervalued (+Discovery)**
TikTok Creator & Influencer	8.3%	0.8%	5.1%	**6.6%**	**Undervalued (+Discovery)**
Walmart Connect & Retail Media	18.5%	28.9%	22.8%	**23.1%**	**Over-credited by Last-Touch**

5. AI Constrained Non-Linear Budget Optimization

Using non-linear Hill saturation response functions with SciPy SLSQP optimization, the AI model balances diminishing marginal returns against P&G's unit economics. Rather than blindly chasing historical ROAS, the algorithm calculates Marginal ROAS (dRevenue/dSpend) at each channel's operating point to identify remaining headroom before saturation.



P&G Channel	Current Spend	AI Target Spend	Delta ($)	Delta (%)	Current ROAS	Target ROAS	Strategic Recommendation
Amazon Ads (Retail Media)	$2,800,000	$3,002,617	+$202,617	+7.2%	1.55x	**1.52x**	Expand / Scale
Walmart Connect & Retail Media	$2,400,000	$2,632,827	+$232,827	+9.7%	1.5x	**1.49x**	Expand / Scale
Google Shopping & Search	$1,600,000	$2,038,477	+$438,477	+27.4%	1.58x	**1.41x**	Expand / Scale
Meta Social (IG / FB Reels)	$2,100,000	$2,133,066	+$33,066	+1.6%	1.32x	**1.26x**	Maintain / Fine-tune
Connected TV & YouTube Video	$1,200,000	$1,093,559	-$106,441	-8.9%	1.16x	**1.23x**	Cut / Defund
P&G; Good Everyday Rewards CRM	$300,000	$660,000	+$360,000	+120.0%	3.36x	**2.23x**	Expand / Scale
TikTok Creator & Influencer	$500,000	$384,454	-$115,546	-23.1%	1.05x	**1.19x**	Cut / Defund
Programmatic Open-Web Display	$1,100,000	$55,000	-$1,045,000	-95.0%	0.37x	**1.09x**	Cut / Defund

6. P&G Root Cause Diagnostics & Waste Liquidation

The AI audit identifies three primary vectors of capital erosion across P&G's digital deployments:

1. **Programmatic Open-Web Bleed:** $1,100,000/month deployed across third-party ad networks suffers from poor viewability, low purchase intent (0.5% CVR), and high household churn (64%). Recommendation: Defund spend by 85% down to a maintenance floor ($165k/mo) and redirect $935k into verified retail media.
2. **TikTok Creator Campaign Churn:** While delivering strong Gen-Z reach, TikTok conversions yield an average household LTV of only $115 with 46% churn within 90 days. Recommendation: Trim budget by 35% ($175k) and mandate direct retailer checkout links.
3. **Under-invested Retail Media & Loyalty Engines:** Amazon Ads, Walmart Connect, and P&G Good Everyday CRM generate premium repeat purchasers (LTV $340 - $420, payback < 2.5 months), yet were artificially constrained by historical channel caps.

7. Procter & Gamble Brand Governance & Phased Implementation Roadmap

Implementation Phase	Target Operational Action	Capital Reallocation	Target P&G Business Impact
Phase 1: Week 1 - 2 Immediate Ad-Waste Liquidation	Slash Programmatic Open-Web Display by $935,000/mo. Restrict TikTok influencer budgets to verified retail conversion campaigns.	-$1,040,750 / month saved from bleed channels	Immediate cessation of negative-ROI display spend; protects $3.1M quarterly cashflow.
Phase 2: Week 3 - 6 Scale Retail Media & Loyalty	Redeploy liberated capital into Amazon Ads Sponsored Products (+$450k), Walmart Connect (+$350k), and P&G Good Everyday CRM (+$180k).	+$980,000 / month redeployed to scalable channels	Portfolio ROAS expands from 1.38x to 1.45x; +$839,855/mo incremental gross revenue.
Phase 3: Month 2 - 3 Autonomous AI Reallocation	Integrate automated API bid adjusters with Amazon Marketing Cloud and Walmart Luminate. Recalibrate SLSQP solvers weekly.	Autonomous weekly spend balancing	Dynamic equilibrium ensuring portfolio household LTV:CAC remains above 3.0x across all brands.

8. Technical Architecture & Mathematical Formulation

• **Markov Chain Removal Attribution:** Directed state graph G=(V, E) calculating incremental removal effect matrices R_i = (P_{conv} - P_{conv \setminus i}) / P_{conv} across 5,000 multi-touch household journeys.
• **Hill Saturation Response Model:** Revenue R_i(S_i) = K_i \cdot (S_i^{n_i}) / (S_{half,i}^{n_i} + S_i^{n_i}) estimating channel diminishing returns and numerical marginal ROAS derivatives.
• **Non-Linear Constrained Solver:** SciPy SLSQP maximizing Net Contribution Margin subject to Sum(S_i) = B_{total} ($12M) and channel liquidity bounds L_i <= S_i <= U_i.
• **Enterprise Decision Intelligence:** Direct integration into P&G Retail Media APIs and First-Party Consumer Data Platforms (CDP).

Report Generated For: Procter & Gamble (P&G) Brand Leadership	**Recommendation Status: APPROVED FOR DEPLOYMENT**
Verified By: AI Decision Intelligence & Autonomous Operations Lab	**Next Review Cycle: 30 Days Post-Reallocation**